Examiner only

4. Beetroot contains a red pigment called betalain, which is stored within the cell vacuole. When the cell membranes are disrupted the pigment leaks out into the surrounding solution, colouring it red.

Some students investigated the permeability of the membranes of beetroot cells. A colorimeter was used to measure the absorbance of the surrounding solution. The absorbance of the solution is proportional to the concentration of betalain.

The students made the following prediction:

"The absorbance recorded will be directly proportional to the temperature of incubation."

The experiment was carried out as follows:

- 6 cubes of fresh beetroot were washed in cold water for 1 minute.
- The cubes were transferred into separate boiling tubes, each containing 20 cm^3 of water.
- The boiling tubes were incubated for 10 minutes in beakers at different temperatures which were maintained with the addition of hot or cold water.
- 3 cm^3 of water from around the beetroot cubes was removed and a colorimeter was used to measure the absorbance, at 540 nm, of each solution.

The experiment was repeated three times, the results are shown in **Table 4.1**.

Table 4.1

Incubation temperature / °C	Absorbance at 540nm / a.u.			
	Experiment 1	Experiment 2	Experiment 3	Mean
10	12	15	21	16
30	15	20	19	18
40	18	22	32	24
60	51	68	71	63
70	95	96	97	96
80	98	98	98	98

(a) (i) State which temperature provided the most reliable results and use your knowledge of cell structure to explain why this might be the case. [2]

....

....

....

....

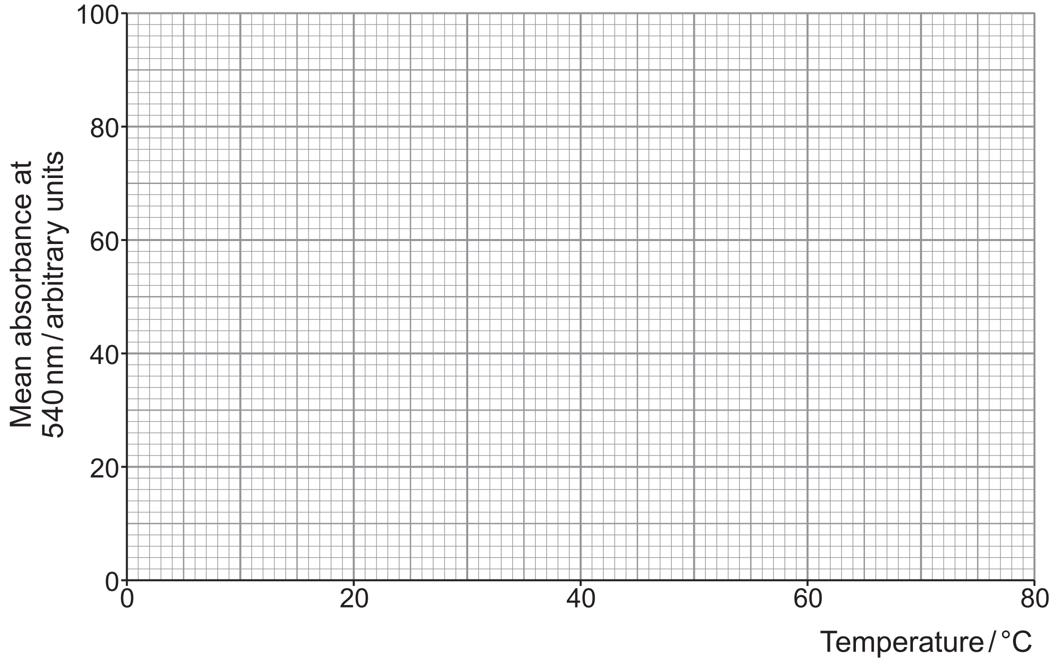
....

....

Examiner only

(ii) Plot the mean results from **Table 4.1** on **Graph 4.2**. [3]

Graph 4.2



(iii) Explain why the students' prediction was not entirely correct. [1]

....

....

....

(iv) Use your knowledge of cell membrane structure to explain the shape of **Graph 4.2**. [2]

....

....

....

....

....

....

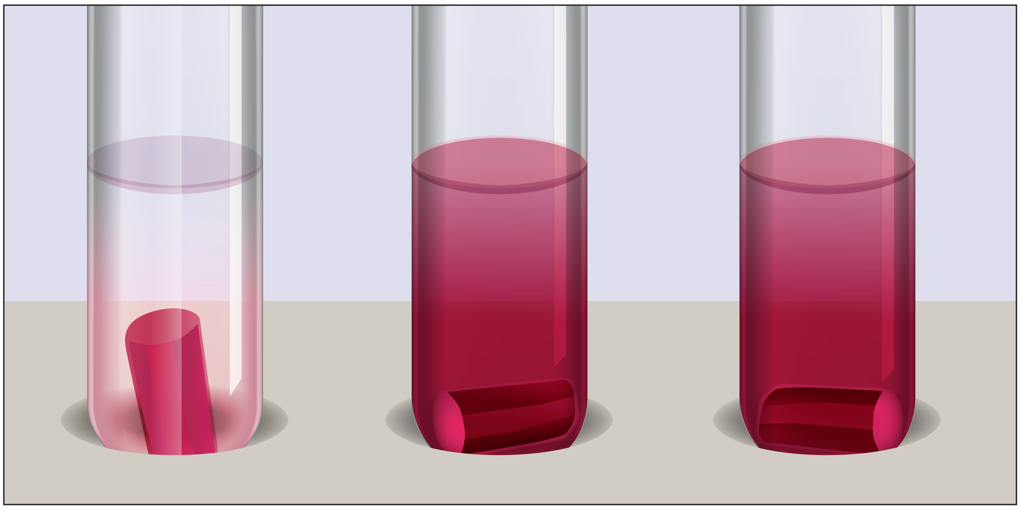
....

....

Examiner only

(b) Another group of students investigated how the type of surrounding solution affected the permeability of membranes in beetroot. They tested water, acid and ethanol. **Image 4.3** shows the experiment after incubation at 20 °C for 10 minutes.

Image 4.3



(i) Using your knowledge of the structure of membranes, conclude how both **acid** and **ethanol** affect the membrane and explain the appearance of the solution in these tubes. [3]

....

....

....

....

....

....

(ii) The students also noted that the beetroot in the water had increased in size. Explain this observation. [3]

....

....

....

....

....

....

14